5. Example (NO_2 pixel)

Assume:

- VCD = 5×10^{15} molecules/cm²
- pixel = 3 km × 3 km = 9×10^{6} m²

Step 1: convert area

$$9 \times 10^{6}\, m^2 = 9 \times 10^{10}\, cm^2$$

Step 2: molecules in pixel

$$N = 5 \times 10^{15} \times 9 \times 10^{10} = 4.5 \times 10^{26}\, molecules$$

Step 3: convert to mass

$$M = \frac{4.5 \times 10^{26}}{6.022 \times 10^{23}} \times 46 \approx 3435\, g$$

~3.4 kg NO_2 in that pixel column